

Quantum LDPC Codes

Lecture 1

Authors: Chirag Raisingh, Kilian Poirier

Lecture Outline

- Classical LDPC codes
 - Reminder on classical codes
 - Tanner graphs
- Quantum LDPC codes
 - Encoding rate problem
 - Good quantum codes
 - Limitations on local codes
 - qLDPC codes constructions
 - Hypergraph Product codes
 - Bivariate Bicycle codes
 - Lifted product codes

Classical LDPC codes

Classical codes

Reminder: We defined classical codes from the generator matrix G defined from the codewords x_i

$$G = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix}$$

We can also refer to the code using its parity check matrix H such that

$$G^T H = H^T G = 0$$

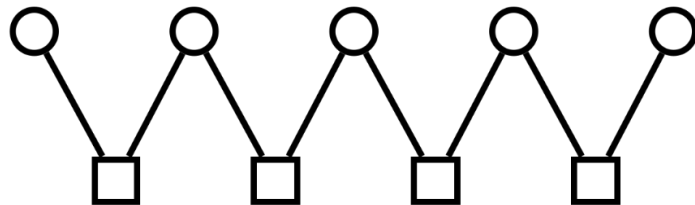
Tanner graphs: Definition

Definition (Tanner graphs). *Given a parity check matrix $H = (h_{ij})_{m \times n}$, the Tanner graph $\mathcal{T} = \mathcal{T}(H)$ is a bipartite graph $\mathcal{T} = (V, C, E)$. Vertices are constructed from the variables nodes $V = \{v_i\}_n$ defined on the columns of H and the check nodes $C = \{c_j\}_m$ defined on the rows of H . Edges connect v_i to c_j if $h_{ij} = 1$.*

Example repetition code:

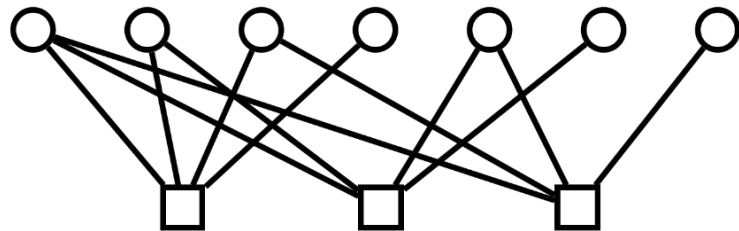
Tanner graphs: Examples

$$H_{\text{rep}} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$



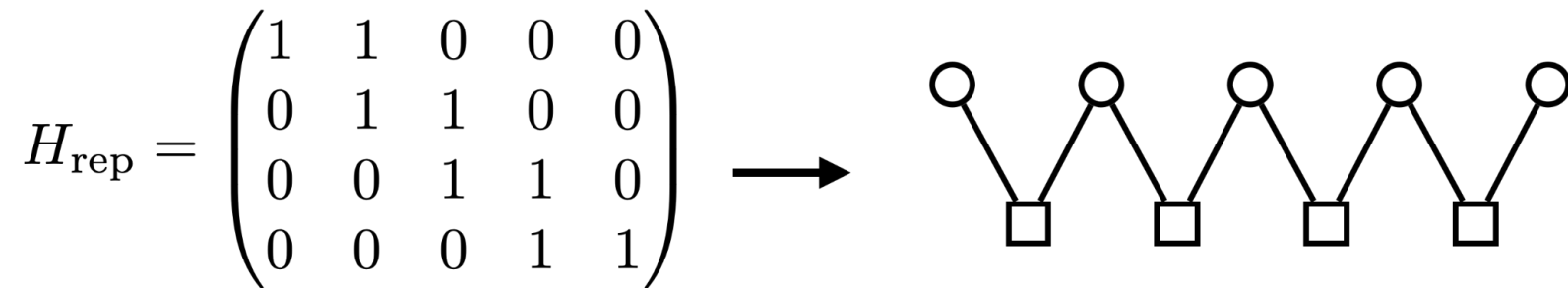
Tanner graphs: Examples

$$H_{\text{Ham}} = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} \longrightarrow$$



LDPC codes

Definition (LDPC codes). A classical binary code $\mathcal{C}$ with parameters $[n, k, d]$ is called a Low Density Parity Check code if its parity check matrix H is w -limited (each row and column has weight at most w), where $w = O(1)$.



Good classical codes

A “good” code means $k = \Theta(n)$, $d = \Theta(n)$

Classically finding codes with linear rate and linear distance is easy (probabilistic method)

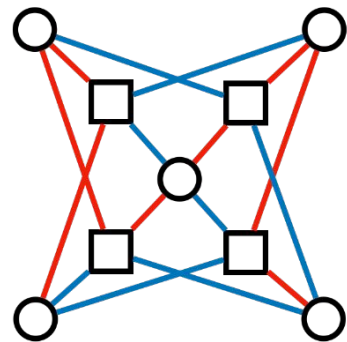
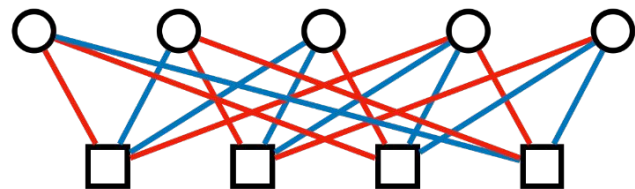
Quantum LDPC codes

Quantum Tanner graph: Definition

Similarly to the classical case, we can define a quantum Tanner graph from a Tanner graph with an additional label for the edges X, Y, Z

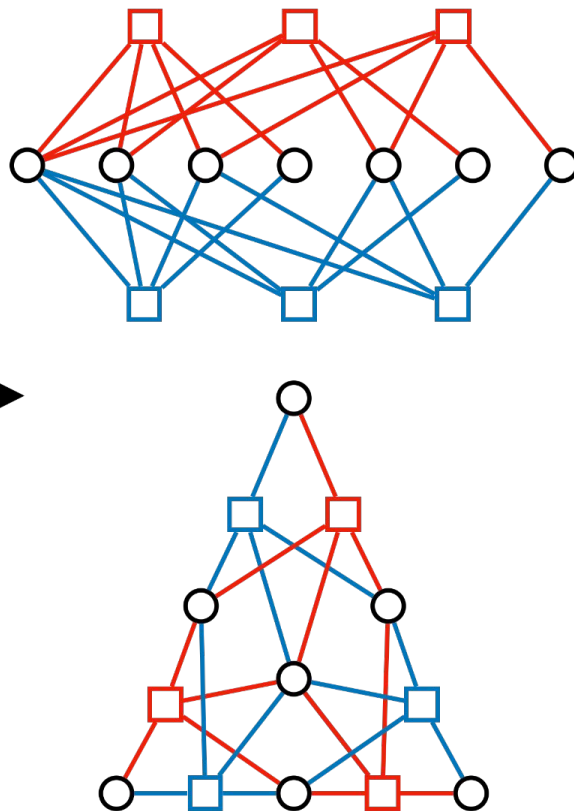
Quantum Tanner graph: Examples

$$H_{5q} = \left(\begin{array}{ccccc|ccccc} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$



Quantum Tanner graph: Examples

$$H_{\text{Steane}} = \left(\begin{array}{cccccc|cccccccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{array} \right) \longrightarrow$$



Quantum LDPC codes

Definition (Quantum LDPC codes). *A CSS quantum code $\mathcal{Q}$ with parameters $[[n, k, d]]$ is called a quantum Low Density Parity Check code if both parity check matrices H_X and H_Z is w -limited (each row and column has weight at most w), where $w = O(1)$.*

Quantum LDPC codes

Definition (Quantum LDPC codes). *A CSS quantum code $\mathcal{Q}$ with parameters $[[n, k, d]]$ is called a quantum Low Density Parity Check code if both parity check matrices H_X and H_Z is w -limited (each row and column has weight at most w), where $w = O(1)$.*

- Repetition code: $[[O(n), 1, d_{X/Z} = O(n)]]$
- Steane code: $[[7, 1, 3]]$
- Five qubit code: $[[5, 1, 3]]$
- Surface code: $[[O(n), 1, O(\sqrt{n})]]$
- Toric code: $[[O(n), 2, O(\sqrt{n})]]$

Good Stabilizer Codes

Unlike classical codes, finding good stabilizer codes is non-trivial

Good Stabilizer Codes

Theorem (Gilbert-Varshamov). *There exists a sequence of $[[n, k, d]]$ stabilizer codes with $n \rightarrow \infty$ and k, d linear in n .*

More precisely, for all $\delta \in [0, 1]$ and for all $\epsilon > 0$, we can achieve:

$$\begin{aligned}\frac{k}{n} &= 1 - h(\delta) - \delta \log_2 3 - \epsilon \\ \frac{d}{n} &\geq \delta,\end{aligned}$$

where $h(\delta) = -\delta \log_2 \delta - (1 - \delta) \log_2 (1 - \delta)$

Good stabilizer codes: Proof

Proof outline:

$Y_{\delta n} :=$ number of Pauli errors $E \in \mathcal{P}_n \setminus S$ with $\sigma(E) = 0$ with weight $|E| \leq \delta n$

- 1- Show that if $\frac{k}{n}$ is small enough, then $\mathbb{E}(Y_{\delta n}) \rightarrow 0$ when $n \rightarrow \infty$
- 2- This proves the existence of a family of codes with $Y_{\delta n}(\mathcal{C}) = 0$
- 3- By definition of $Y_{\delta n}$, this shows that $d > \delta n$

Good stabilizer codes: Proof

How many stabilizer codes $[[n, k]]$?

Good stabilizer codes: Proof

How many stabilizer codes $[[n, k]]$ such that $\sigma(E) = 0$?

Good stabilizer codes: Proof

$Y_{\delta n} :=$ number of Pauli errors $E \in \mathcal{P}_n \setminus S$ with $\sigma(E) = 0$ with weight $|E| \leq \delta n$

Write $Y_{\delta n}$ as:

Good stabilizer codes: Proof

$Y_{\delta n} :=$ number of Pauli errors $E \in \mathcal{P}_n \setminus S$ with $\sigma(E) = 0$ with weight $|E| \leq \delta n$

We just computed this:

Good stabilizer codes: Proof

Summing over all possible errors:

Converges when:

This gives the upper bound:

Good stabilizer codes: Proof

Concludes the proof:

$Y_{\delta n} :=$ number of Pauli errors $E \in \mathcal{P}_n \setminus S$ with $\sigma(E) = 0$ with weight $|E| \leq \delta n$

- 1- Show that if $\frac{k}{n}$ is small enough, then $\mathbb{E}(Y_{\delta n}) \rightarrow 0$ when $n \rightarrow \infty$
- 2- This proves the existence of a family of codes with $Y_{\delta n}(\mathcal{C}) = 0$
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Local codes

Theorem (Bravyi-Poulin-Terhal bound). *Qubit stabilizer codes defined on D -dimensional Euclidean lattices are limited by*

$$d = O(n^{1-1/D}),$$

and

$$kd^{2/(D-1)} = O(n).$$

Hypergraph Product codes

Hypergraph Product (HP) codes: Definition

Given two codes as Tanner graphs $\mathcal{G}_1 = \mathcal{T}(V_1, C_1, E_1)$, $\mathcal{G}_2 = \mathcal{T}(V_2, C_2, E_2)$

We define the hypergraph product code as the CSS code constructed from the vertex set:

$$V := V_1 \times V_2, C_1 \times C_2$$

and the set of checks:

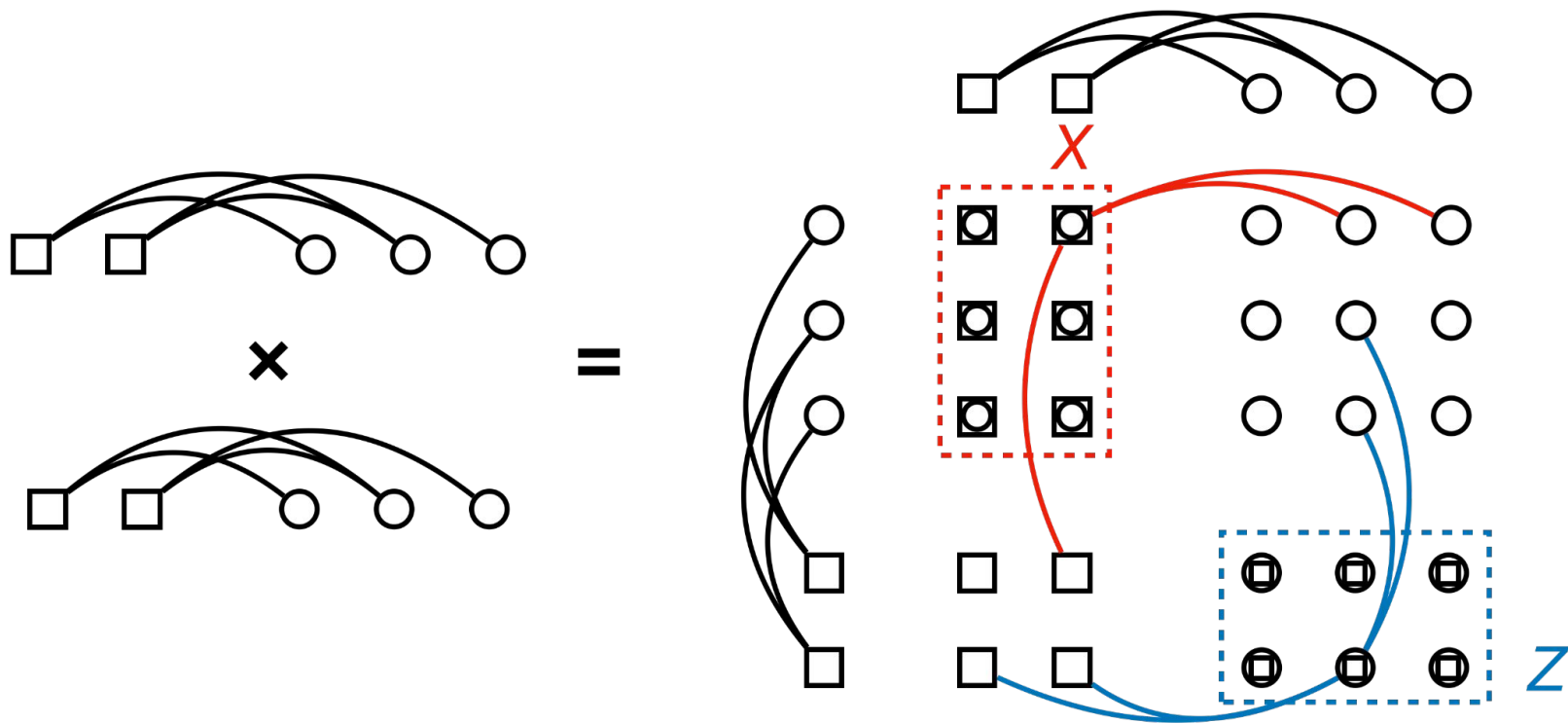
$$C := C_1 \times V_2, V_1 \times C_2$$

Edges are defined from connected components.

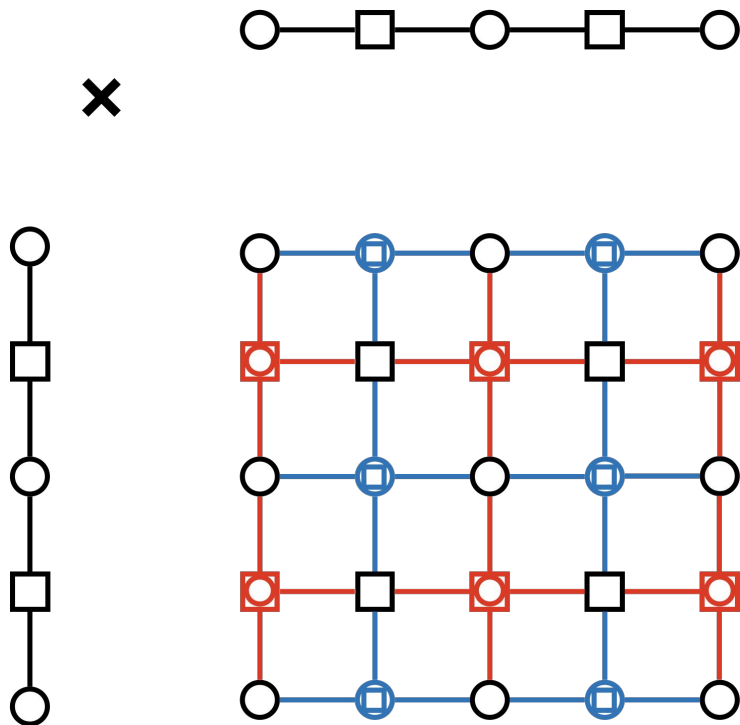
Hypergraph Product (HP) codes: Construction

From two repetition codes:

Hypergraph Product (HP) codes: Example



Hypergraph Product (HP) codes: Example



Hypergraph Product (HP) codes: Properties

We can construct the CSS code from its parity check matrices

$$H_X = (H_1 \otimes I_{n_2} \quad I_{r_1} \otimes H_2^T) \quad H_Z = (I_{n_1} \otimes H_2 \quad H_1^T \otimes I_{r_2})$$

The resulting code $HP(\mathcal{G}_1, \mathcal{G}_2)$ has parameters:

$$n = n_1 n_2 + r_1 r_2$$

$$k = k_1 k_2$$

$$d = \min(d_1, d_2)$$

We just created a $[[n, O(n), O(\sqrt{n})]]$ code

Bivariate Bicycle codes

Bivariate Bicycle (BB) codes

Definition (Bivariate Bicycle (BB) code). *A BB code is defined by the pair of matrices*

$$A = A_1 + A_2 + A_3 \quad \text{and} \quad B = B_1 + B_2 + B_3$$

where each matrix A_i and B_j is a power of x or y . We also assume that A_i and B_j are distinct to avoid cancellation of terms. Note that both A and B have exactly three nonzero entries in each row and each column.

We can define a BB code as $BB(A, B)$ with block length $n = 2lm$ and check matrices

$$H_X = [A \ B] \quad \text{and} \quad H_Z = [B^T \ A^T].$$

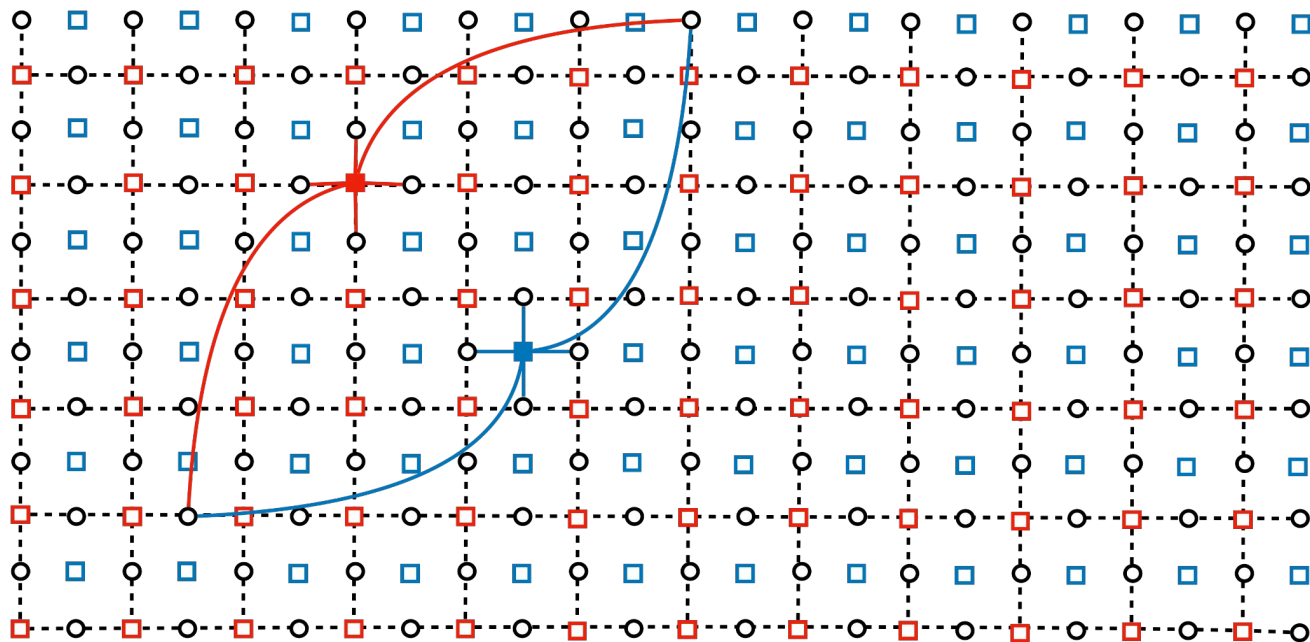
$$x = S_l \otimes I_m \quad y = I_l \otimes S_m \quad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad S_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Bicycle Bivariate (BB) codes: $\llbracket 144, 12, 12 \rrbracket$

Choosing $l = 12$, $m = 6$ and $A = x^3 + y + y^2$, $B = y^3 + x + x^2$

Bicycle Bivariate (BB) codes: $[[144, 12, 12]]$

Choosing $l = 12$, $m = 6$ and $A = x^3 + y + y^2$, $B = y^3 + x + x^2$



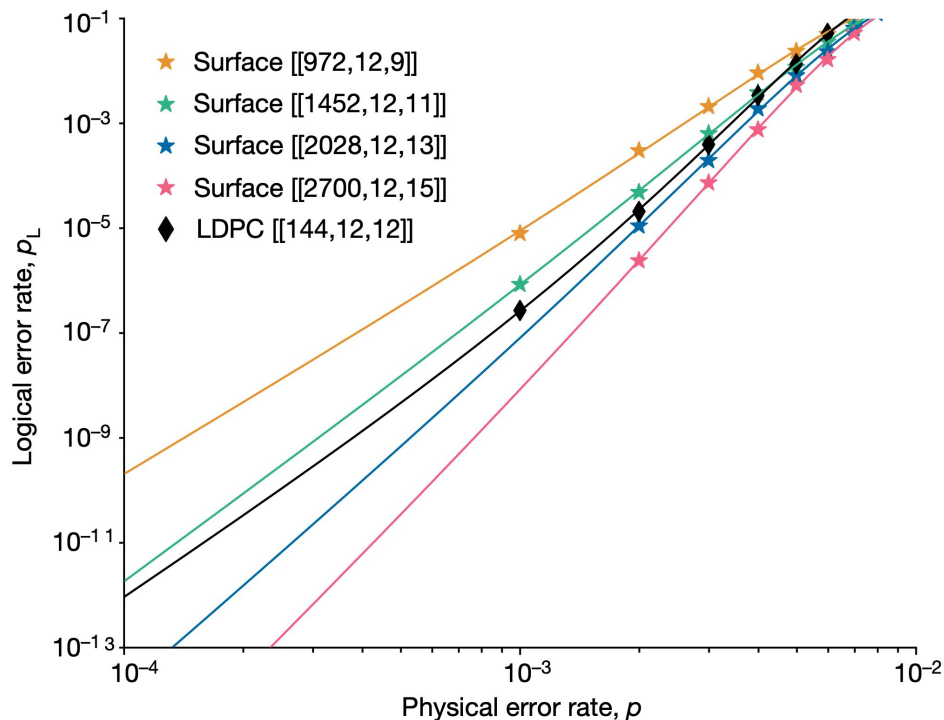
Bicycle Bivariate (BB) codes: $[[144, 12, 12]]$

Non-local checks: weight 6

High effective encoding rate: $k = 1/24$

High threshold: $p_0 = 0.0065$

High effective distance: $d_{\text{circ}} \leq 10$



Lifted Product codes

Lifted Product (LP) codes: Definition

Definition (Lifted Product (LP) code). *A LP code is defined by the pair of matrices $A \in \mathcal{M}_{m_A \times n_A}(R)$ and $B \in \mathcal{M}_{m_B \times n_B}(R)$ over the ring R . $LP(A, B)$ is described by the parity check matrices as*

$$\begin{aligned} H_X &= (A \otimes I_{m_B} \quad I_{m_A} \otimes B) \\ H_Z &= (I_{n_A} \otimes B^* \quad A^* \otimes I_{n_B}). \end{aligned}$$

$$A^* = A(p(x) \rightarrow p(x^{-1}))^T$$

Lifted Product (LP) codes: Properties

The coefficients of the matrices are lifted from $\mathbb{F}_2$ to the elements of the ring R

This class of codes can achieve $\llbracket n, k = \Theta(n), d = \Theta(n / \log n) \rrbracket$ for good choices of parameters and ring.

Hypergraph product codes are a special case where $R = \mathbb{F}_2 \cong \mathcal{M}_1(\mathbb{F}_2)$

BB codes are special cases where defined on the abelian group $\mathbb{Z}_l \times \mathbb{Z}_m$

Lifted Product (LP) codes: Example

Hypergraph product codes are a special case where

Choosing the ring to be $R := \mathbb{F}_2[x]/(x^l + 1)$ we can find the code $LP(A, A^*)$ with parameters:

$$\llbracket n = (r_A^2 + n_A^2)l, \ k \geq (n_A - r_A)^2l, \ d \leq \min((r_A + 1)!, (n_A + 1)!) \rrbracket$$

Example: $\mathbf{lp}_{20}^{3,5}$

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & x^{14} & x^{19} & x^{11} & x^{26} \\ 1 & x^{13} & x^2 & x^{15} & x^{21} \end{pmatrix} \quad \llbracket 1122, 148, d \leq 20 \rrbracket$$

Summary

- Tanner graphs: Classical and Quantum codes
- LDPC and qLDPC codes
 - Good codes, local codes
 - Encoding rate problem
- qLDPC codes constructions
 - Hypergraph Product codes
 - Bivariate Bicycle codes
 - Lifted product codes